

Physical AI SDK: 2026 Model Lifecycle & Delivery Strategy

What We're Delivering

Structured 4-Phase Model Lifecycle:

ENABLEMENT → BENCHMARKING → PROFILING → FINE-TUNING
(1 wk) (3 days) (3 days) (2-3 wk)

2026 Portfolio Targets:

- **20 production-ready models** across various categories
- **3 market segments:** Robotics, Healthcare, Industrial
- Unified SDK installer with integrated tooling

Model Categories:

Category	Models	Status
Vision/Detection	YoloV12, MobileSAM, CrossFormer,...	Active
VLMs	Qwen2.5-VL, Qwen3-VL, LLaMA3.2-Vision,...	Active
LLMs	DeepSeek-R1,...	Not started
Speech	Whisper (ASR), XTTS (TTS),...	Active
3D/Robotics	CenterPoint, OpenVLA,...	In Progress
Medical	MedSigLIP, 3D U-NET,...	Not started
...	...	...

2026 Release Roadmap

R1: Foundation

- Lifecycle framework operational
- CI/CD automation & GitHub Pages/AMD docs
- Release logistics streamlined

R2: Scale

- **8 production-ready models** (P0/P1)
- Single SDK installer
- Quantization & Infra Beta (vLLM, ONNX-RT, Llama.cpp)
- Sample apps: ChatBot, Object Detection, GStreamer+ROCM

R3: Expand

- **8 additional models** (16 total)
- Enhanced quantization & profiling tools
- Robotics simulator integration
- Model deployment infrastructure

R4: Complete 1st cycle

- **20 total production models**
- Multi-segment deployment ready
- Model training infrastructure (Drafter)
- 2027 strategy planning

Open Source Release & Distribution Strategy

HuggingFace Hub Distribution

What We Publish:

- 🤗 **Pre-optimized Model Weights**: ONNX, TensorRT, ROCm-ready
- 📦 **Quantized Variants**: INT8, FP16 for edge deployment
- 🐳 **Docker Images**: GPU/NPU/Hybrid ready-to-run containers
- 📊 **Benchmark Results**: Verified performance metrics included
- 📄 **Model Cards**: Documentation, usage examples, hardware requirements

HuggingFace Advantages:

- ✅ **Public Discovery**: Searchable by AI/robotics community
- ✅ **Version Control**: Track model iterations & improvements
- ✅ **Direct Download**: `pip install` or one-click deployment
- ✅ **Community Engagement**: Issues, feedback, contributions
- ✅ **Ecosystem Integration**: Works with Transformers, ONNX-RT, vLLM

Infrastructure Tools (Open Source Ready):

- **vLLM**: Efficient LLM serving & inference
- **Llama.cpp**: Optimized CPU/GPU inference
- **ONNX Runtime**: Cross-platform model deployment
- **Lemonade Profiler**: Performance analysis tool